

Transportation Problem
Modified Distribution Method (MODI)
Initial Solution using Vogel's Approximation Method

	1	2	3	4	5	Supply
A	²⁰ 40	²⁰ 25	⁶⁰ 22	33	0	100
B	44	35	30	30	³⁰ 0	30
C	²⁰ 38	38	28	³⁰ 30	²⁰ 0	70
Demand	40	20	60	30	50	200

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	1	2	3	4	5	Supply	
A	40 46	20 25	60 22	33 33	20 0	100	$u_A = 0$
B	44	35	30	30	50 0	30	$u_B = -2$
C	40 38	38	28	30 30	0 0	70	$u_C = -2$
Demand	40	20	60	30	50	200	
	v_1 40	v_2 25	v_3 22	v_4 32	v_5 2		

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Step 1: Check for Degeneracy

	1	2	3	4	5	Supply	
A	40	²⁰ 25	⁶⁰ 22	33	²⁰ 0	100	
B	44	35	30	30	⁵⁰ 0	30	
C	⁴⁰ 38	38	28	³⁰ 30	0	70	
Demand	40	20	60	30	50	200	

$$\begin{aligned} \text{Allocations} &= 6 \\ 3 + 5 - 1 &= 7 \end{aligned}$$

$$6 \neq 7$$

Solution is Degenerate

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Step 1a: Resolving Degeneracy

Solution is Degenerate.

We must add a small allocation ϵ to
unallocated cell with the minimum

	1	2	3	4	5	Supply	
A	40	²⁰ 25	⁶⁰ 22	33	²⁰ 0	100	
B	44	35	30	30	⁵⁰ 0	30	
C	⁴⁰ 38	38	28	³⁰ 30	ϵ 0	70	
Demand	40	20	60	30	50	200	

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Step 2: Calculate u_i and v_j

Use $C_{ij} = u_i + v_j$ for all allocated cells.

Start by setting $u_A = 0$

	1	2	3	4	5	Supply	
A	40	²⁰ 25	⁶⁰ 22	33	²⁰ 0	100	$u_A = 0$
B	44	35	30	30	⁵⁰ 0	30	$u_B = 0$
C	⁴⁰ 38	38	28	³⁰ 30	[€] 0	70	$u_C = 0$
Demand	40	20	60	30	50	200	
	v_1 38	v_2 25	v_3 22	v_4 30	v_5 0		

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Step 3: Calculate Opportunity Costs (d_{ij})

For all unallocated (white) cells, use:

$$d_{ij} = u_i + v_j - C_{ij}$$

Opportunity Costs for unallocated cells.

	1	2	3	4	5	Supply	
A	40	20 25	60 22	33	20 0	100	$u_A = 0$
B	44	35	30	30	50 0	30	$u_B = 0$
C	40 38	38	28	30 30	€ 0	70	$u_C = 0$
Demand	40	20	60	30	50	200	
	v_1 38	v_2 25	v_3 22	v_4 30	v_5 0		

$$d_{A1} = -2$$

$$d_{A4} = -3$$

$$d_{A5} = -6$$

$$d_{B2} = -10$$

$$d_{B3} = -8$$

$$d_{B4} = 0$$

$$d_{C2} = -13$$

$$d_{C3} = -6$$

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Step 4: Check for Optimality

Optimality Condition: All $d_{ij} \leq 0$

Opportunity Costs for unallocated cells.

	1	2	3	4	5	Supply
A	40	²⁰ 25	⁶⁰ 22	33	²⁰ 0	100
B	44	35	30	30	⁵⁰ 0	30
C	⁴⁰ 38	38	28	³⁰ 30	[€] 0	70
Demand	40	20	60	30	50	200
	v_1	v_2	v_3	v_4	v_5	
	$\parallel$	$\parallel$	$\parallel$	$\parallel$	$\parallel$	
	38	25	22	30	0	

$$d_{A1} = -2$$

$$d_{A4} = -3$$

$$d_{A5} = -6$$

$$d_{B2} = -10$$

$$d_{B3} = -8$$

$$d_{C2} = -13$$

$$d_{C3} = -6$$

$$d_{C4} = -6$$

$$d_{C5} = -6$$

All opportunity costs are ≤ 0 .

The solution is OPTIMAL.

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Step 4: Check for Optimality

Optimality Condition: All $d_{ij} \leq 0$

Opportunity Costs for unallocated cells.

$$d_{A4} = -1$$

$$d_{A5} = -2$$

$$d_{B1} = -6$$

$$d_{B2} = -12$$

$$d_{B3} = -10$$

$$d_{C1} = -2$$

$$d_{C4} = 0$$

$$d_{C2} = -15$$

$$d_{C3} = -8$$

At least one opportunity cost is > 0 .

The solution is NOT optimal.

Select cell with most positive d_{ij} as entering cell.

	1	2	3	4	5	Supply	
A	20 40	20 25	60 22	33	0	100	$u_A = -6$
B	44	35	30	30	30 0	30	$u_B = -2$
C	20 38	38	28	30 30	20 0	70	$u_C = -4$
Demand	40	20	60	30	50	200	
	v_1 40	v_2 25	v_3 22	v_4 32	v_5 2		

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Step 5: Identify Closed Loop

	1	2	3	4	5	Supply	
A	20 40	20 25	60 22	30 33	0 9	100	$u_A = 0$
B	44	35	30	30	30 0	30	$u_B = -2$
C	20 38	38	28	30 36	20 0	70	$u_C = -2$
Demand	40	20	60	30	50	200	
	v_1 40	v_2 25	v_3 22	v_4 32	v_5 2		

Loop Rules:

1. Start at the entering cell.
2. Move horizontally or vertically.
3. Turn **only** at allocated cells.
4. End at the starting cell.

Assign alternating $+/-$ signs, starting with $+$ at the entering cell.